

Raunak Kumar

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<https://raunakkmr.github.io>

SUMMARY

I am passionate about algorithms, machine learning, systems and competitive programming. I take keen interest in machine learning research and teaching.

EDUCATION

B.Sc. Computer Science,

September 2013 - May 2018

University of British Columbia, Vancouver, BC

- Grade in Computer Science and Mathematics courses: 90.4%
- Related Coursework:
 - Machine Learning (graduate course), Advanced Algorithm Design and Analysis, Randomized Big Data Algorithms (graduate course), Theory of Computation
 - Probability, Real Analysis, Graph Theory, Linear Algebra, Linear Programming
 - Internet Computing, Advanced Operating Systems, Databases

AWARDS

J. Fred Muir Memorial Scholarship in Science ('13-'14)

Trek Excellence Scholarship ('13-'14) - Top 5% of undergraduates

Science Scholar / Dean's Honors List ('13-'14, '14-'15, '15-'16, '16-'17)

UBC Computer Science Teaching Assistant Award ('16-'17)

TECHNOLOGY SKILLS

Programming Languages: C++, Python, C, Matlab, Java.

Software: TensorFlow, PyTorch, Numpy, Scikit-Learn, Vim, Git.

Operating Systems: Linux Ubuntu, Windows.

EXPERIENCE

Software Engineering Intern

May 2016 - August 2016

Google, Mountain View, CA

- Worked on a distributed datastore trigger utility in C++ for the App and Product Infrastructure team. Triggers are a mechanism by which actions can be scheduled for the future. The goal of the utility is to provide an easy and extensible way to maintain and debug triggers.

NSERC Research Assistantship

May 2017 - Present

Computer Science Department, UBC

- Working on theoretical and applied machine learning problems under the supervision of Prof. Mark Schmidt. We give the first convergence rate for non-convex functions in the surrogate optimization framework. We view expectation-maximization (EM) in this framework and, consequently, also provide a convergence rate for EM. We are also working on using other ideas from optimization to improve EM. Along with a graduate student, I am also working on improving deep learning architectures for image segmentation.

TEACHING

Instructor / Coordinator

January 2017 - April 2017

Computer Science Department, UBC

- Co-taught CPSC 490: Problem Solving in Computer Science, focusing on practical applications of advanced algorithms and data structures. Responsibilities included lecturing a class of 15-30 students 3 times a week, writing lecture notes and writing assignments. Website: www.ugrad.cs.ubc.ca/~cs490/2016W2/.

Teaching Assistant

September 2014 - Present

Computer Science Department, UBC

- Worked as a TA for 8 terms for the following courses: Advanced Algorithm Design and Analysis, Machine Learning, Advanced Operating Systems, Intermediate Algorithm Design and Analysis, Introduction to Computer Systems, Models of Computation. Responsibilities included lecturing sections ranging from 15 - 100 students, holding office hours and grading. I have taught an entire lecture for two algorithms courses as well. I was an undergrad when I worked as a TA for the Advanced Algorithms and Machine Learning courses. Usually, only grad students are assigned to TA these courses.

PROJECTS

Programming Competitions

- UBC ACM Team: 2015 - Present.
- UBC ACM Team Coach (Div. 2) - 2017 - Present.
- ACM-ICPC PacNW Regionals: 15th (2016, Div. 1), 6th (2015, Div. 2).
- Microsoft Coding Competition MSFT3C: 2nd (2017), 6th (2016).
- SFU Winter Programming Contest: 8th (2017).
- NAIPC (Open Division USA/Canada): 27th (2016).
- Calgary Collegiate Programming Competition: 3rd (2015, Div. 2).

Approximating Steiner Trees in the Semi-Streaming Model

- Presented a $2(1+\epsilon)(2^{\frac{\log n}{\log \log n}} + 1)$ -approximation for the weighed metric Steiner tree problem using $O(n \text{ polylog } n)$ space, as part of a course project for a graduate class on randomized big data algorithms. Paper can be found on my website.

Machine Learning Reading Group

- I present at the Machine Learning Reading Group and the Machine Learning Theory Reading Group. I have presented on topics on learning theory, reinforcement learning and online learning. Slides can be found on my website.

Gas Detection

- Implemented random forest and neural network classification to detect dangerous gases based on sensor measurements for a company in Victoria, BC. Also implemented visualization of the training data using PCA and ISOMAP.

Kernel

- Implemented basic functionality of a kernel using C and x86 assembly on the Bochs emulator. Features included memory management, process management, system calls, interrupt handling, signals and hardware interrupts.

Web Proxy Server

- Created a multithreaded server application using C and Unix sockets to access HTTP content from websites, block blacklisted websites and cache results.

DNS Client

- Created a scaled down DNS resolver using Java and UDP datagrams which interacts with various DNS servers to resolve a domain name into an IP address.

Dictionary Client

- Created a client side application using Java and TCP to connect with specified dictionary servers and retrieve the requested definitions.

insightUBC

- Created a backed using Typescript and Node.js to query UBC rooms and courses data.

REFERENCES

William Evans, email: will@cs.ubc.ca, phone: 604-822-0827
Mark Schmidt, email: schmidt@cs.ubc.ca, phone: 604-822-6421
Nicholas Harvey, email: nickhar@cs.ubc.ca, phone: 604-822-5491

OTHER

Canadian citizen.